

From FSD to FSC: Enabling Full Smart-Communication in Autonomous Vehicles Through Full Self-Driving Models

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Abstract—In-vehicle wireless systems, such as vehicle-to-satellite communication, face challenges in adapting to rapidly changing environments due to the inherent latency of aligning high-precision beamforming with distant base stations (e.g., satellites). Motivated by the robust sensing capabilities of today's autonomous vehicles, we propose modeling the radio environment in parallel. This allows the in-vehicle communication subsystem to predict channel conditions along the vehicle's planned trajectory and precompute optimal beamforming parameters in advance, effectively addressing rapid environmental changes. To this end, we introduce Full Smart-Communication (FSC), a novel framework that repurposes intermediate feature representations—such as planned trajectories, occupancy maps, and posture estimates—from a vehicle's end-to-end Full Self-Driving (FSD) model to derive spatio-temporal stochastic radio radiance fields. This approach avoids direct access to raw sensor streams while effectively correlating kinematic and environmental states with radio propagation dynamics. Our comprehensive experiments demonstrate that FSC achieves a mean gain of 15 dB and maintains less than one-degree beamforming error at direct upcoming positions. Additionally, FSC reduces power consumption by 70% and GPU memory usage by 94% through the hitchhiking approach. By leveraging FSD-derived environmental awareness for communication, our work bridges the gap between autonomous navigation and robust connectivity, unlocking the synergy of perception and radio resource management in next-generation self-driving vehicles.

I. INTRODUCTION

Modern electric vehicles increasingly demand robust wireless connectivity to support advanced functionalities that extend far beyond basic transportation. High-speed communication links are critical for enabling mission-critical services such as over-the-air (OTA) software updates, real-time traffic and navigation data, and cloud-based diagnostics, as well as occupant-centric applications like multimedia streaming and voice assistants. Emerging paradigms such as vehicle-to-everything (V2X) networks further amplify these requirements, necessitating ultra-reliable, low-latency communication (URLLC) to ensure safety-critical coordination between vehicles, infrastructure, and satellites.

Maintaining stable wireless connectivity in mobile environments remains a significant technical challenge, particularly for communication systems that rely on beamforming techniques, such as mmWave and satellite links. Our field meas-

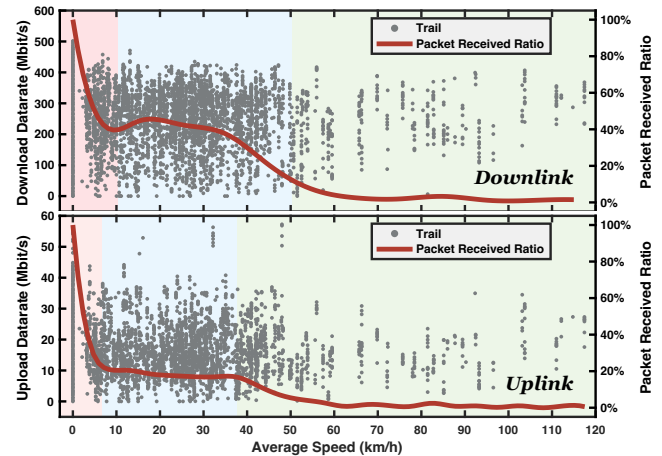


Fig. 1: A V2S communication evaluation on Starlink demonstrates that both downlink and uplink performance exhibit three distinct stages of degradation as vehicle speed increases. When the speed exceeds 40-50 km/h, the communication becomes nearly unavailable.

urements of Starlink in urban settings reveal a 90% throughput decline as vehicle speeds increase from 0 to 50 km/h, as shown in Fig. 1. While this degradation arises from many factors (such as rapid channel fading and Doppler shifts), the primary challenge stems from the dynamic and unpredictable nature of the vehicle's environment. These conditions render historical channel state information obsolete almost instantly, leaving it ineffective for future actions. For instance, abrupt vibrations resulting from road irregularities or obstructions from suddenly emerging structures can significantly misalign phased-array antennas during beamforming toward distant satellites, disrupting link stability.

We observe that modern autonomous vehicles are equipped with advanced multi-sensor fusion systems to support Full Self-Driving (FSD) capabilities. These systems typically integrate six cameras for 360° visual perception, 3~6 LiDARs for high-resolution 3D mapping, and multi-band radars for short- and long-range object detection. Combined with GPS/IMU modules for precise localization, these sensors create a real-time, high-fidelity representation of the vehicle's surroundings. This perception framework not only powers core autonomous functions—such as collision avoidance and path planning—but

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also enables the vehicle to anticipate future environmental states. *Notably, the communication subsystem can leverage this rich sensor data to predict channel conditions along the vehicle's planned trajectory, enabling proactive optimization of wireless connectivity. Surprisingly, this insight has remained overlooked in the past.*

In this work, we introduce Full Smart-Communication (FSC), a framework that leverages the existing sensing infrastructure of autonomous vehicles to model the radio-frequency environment (i.e., radio radiance fields) in which the vehicles operate. By capturing and analyzing this data, FSC predicts optimal channel coefficients for future movements, effectively mitigating the challenge of latency in adapting to rapidly changing environments.

However, implementing FSC from a systematic perspective introduces two primary technical challenges.

- A straightforward approach would be to develop FSC as an independent module operating in parallel with the FSD system, directly accessing sensor data. However, this solution poses significant challenges. Concurrent access to sensors could disrupt or delay the FSD system, introducing unacceptable safety risks. Additionally, processing video streams from six or more cameras would require a dedicated high-performance chip, substantially increasing the cost of communication models.

To achieve cost-efficiency with operational safety, we implemented a hitchhiking approach in the design and execution of the FSC framework. It intercepts the copies of the intermediate results (planned trajectory, occupancy map and posture features) from the FSD system in a passive manner to construct its own scene representations and fine-tune the networks for communication-specific tasks. This methodology circumvents the need for intensive computation typically required for sensing data processing, thereby obviating the necessity for a high-performance, highly-tailored GPU or in-vehicle System on a Chip (SoC). This design also prevents concurrent sensor access, ensuring that the FSD workflow remains uninterrupted and safeguarding the system's operational safety. The details refer to Section IV.

- Second, traditional approaches for developing radio radiance fields, such as NeRF² [1] and RF-Diffusion [2], are designed for static scenes with abundant training samples. They fail to effectively model the radio environment for in-vehicle communications due to the highly dynamic nature of these settings and the scarcity of training data. The rapid environmental changes and limited samples make it difficult for conventional methods to capture the dynamics.

To overcome this challenge, we introduce a progressive ray tracing framework that leverages temporal coherence in dynamic environments. Our approach constructs each frame's radiance field through adaptive reuse and refinement of the preceding frame's volumetric representation. This strategy ensures seamless alignment between consecutive radiance fields while enabling voxels in subsequent frames to inherit previously computed radiance characteristics. By progressively evolving the radiance model across time, our method achieves



Fig. 2: The van equipped with the Starlink FHP receiver. The photo is reshaped from [3].

dual advantages: it minimizes redundant computational overhead while simultaneously bolstering temporal coherence. This synergistic optimization significantly elevates the overall fidelity of radiance field reconstruction in dynamic scenes. The details refer to Section V.

Contributions. Our in-field experiments on a self-driving platform demonstrate that FSC delivers a 15 dB performance gain and maintains a beamforming error of less than 1° compared to conventional in-vehicle V2X systems. Leveraging the hitchhiking paradigm, FSC achieves substantial resource savings—reducing energy consumption by 70.32%, GPU memory usage by 85.56%, and CPU memory usage by 94.42%. Notably, FSC sustains a high signal-to-noise ratio (SNR) of 13 dB or above at speeds up to 50 km/h, whereas Starlink's SNR drops below 5 dB under identical conditions. These results are enabled by a novel hitchhiking paradigm that passively leverages intermediate perception data and progressive ray tracing for efficient, cost-effective, and stable radio radiance field construction without disrupting safety-critical operations. A visual demo is available at <https://youtu.be/ODR212r91ks>.

II. MOTIVATION

Direct-to-satellite communication is emerging as an increasingly significant player in the area of next-generation communication technologies. Its resilience and independence from ground-based infrastructure make V2S communication especially beneficial for vehicles in remote areas, where traditional communication networks are often unavailable. To evaluate how driving affects V2S connectivity, researchers [3] from Osnabrück University conducted field tests with Starlink. As shown in Fig. 2, a Starlink antenna array was mounted on top of a van, and a Raspberry Pi 4 inside periodically connected to the university server when the van traveled at different speeds. Based on the 140,000 collected measurements, we performed the analysis between the speed and the communication quality.

Results. The results are shown in Fig. 1. For the downlink, three distinct stages are observed. In the first stage, the Packet Received Ratio (PRR) decreases sharply as the speed increases from zero to nearly 10 km/h, with the PRR dropping by half. This highlights the significant impact of motion on downlink quality. In the second stage, a slight rebound in PRR occurs between 20–50 km/h, attributed to timely channel adjustments by the communication system. However, in the third stage, when the speed exceeds 50 km/h, the PRR drops sharply from

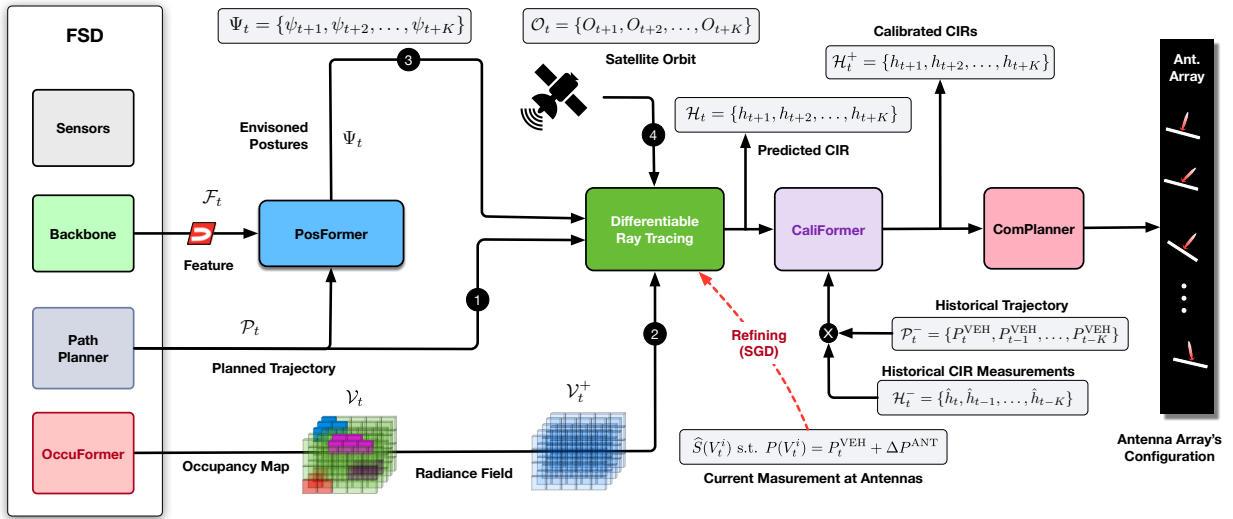


Fig. 3: Architecture of FSC. It generates a new radiance field based on an FSD-derived occupancy map, which is then used for ray tracing. This process is followed by an SGD-based optimization that leverages the current RF measurement obtained from the antenna array. The CIRs along the planned trajectory are retrieved from the radiance field and calibrated for further use.

40% to nearly zero, as the system can no longer keep up with the rapid environmental changes caused by the high speed. A similar trend is observed for the uplink, though the PRR drops more rapidly due to the need for more precise beamforming, making the uplink more sensitive to environmental changes.

Summary. Starlink clients use phased-array antennas to establish directional beams toward satellites, relying on downlink reference signals to estimate channel coefficients and compute beamforming weights for subsequent transmissions. While effective in quasi-static conditions, this approach struggles under vehicular mobility: rapid environmental changes render channel estimates outdated by the next transmission window. At low speeds (<50 km/h), the channel coherence time typically exceeds the beamforming update interval, maintaining sufficient temporal correlation. However, at higher speeds (>50 km/h), beamforming latency makes estimated coefficients obsolete before deployment, creating a mismatch between predicted and actual channel states. *The limitation—conventional V2S systems’ inability to adapt to sub-coherence-time dynamics—motivates our development of a proactive framework that anticipates channel variations through sensor-fused radio environment modeling and learning.*

III. OVERVIEW

Modern autonomous vehicles are equipped with a suite of advanced sensors capable of reconstructing their surrounding environment with centimeter-level accuracy at ranges extending hundreds of meters. This high-resolution spatial awareness inspires our approach to address vehicular communication challenges by leveraging the sensor-derived environmental model. Our proposed FSC framework utilizes real-time outputs from the FSD system, particularly intermediate representations that encode geometric and semantic features of the operational environment.

Fig. 3 illustrates the system architecture of the proposed FSC framework. It captures the planned trajectory $\mathcal{P}_t$, occupancy map $\mathcal{V}_t$, and environmental posture Ψ_t from the FSD model, along with the satellite orbit $\mathcal{O}_t$ obtained from ephemeris data. These inputs are processed by the core component of *differentiable ray tracing* (DRT) to estimate the *radiance field*. As the vehicle moves forward, the radiance field is continuously refined using new RF measurements. To mitigate accumulated errors, a neural component called CaliFormer is designed to calibrate the estimated channel impulse responses (CIRs) $\mathcal{H}_t$. Finally, the calibrated CIRs are fed into the ComPlanner component to generate the beamforming coefficients for the next K steps.

FSC serves as a versatile solution capable of enhancing various forms of wireless communication, including vehicle-to-satellite (V2S), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X), and even 5G communications. In this work, we present FSC using V2S as an example. In the following sections, we elaborate on its design and implementation.

IV. HITCHHIKING ON FSD MODELS

In this section, we introduce the background of the FSD and then details how intermediate data generated by the FSD system is harnessed by our FSC framework.

A. FSD Background

Currently, no full self-driving (FSD) solution has gained broad acceptance in academia or industry. To illustrate the FSD workflow, we highlight two state-of-the-art approaches—UniAD [4] and OccuFormer [5]. As depicted in Fig. 4, typical FSD systems comprise four main modules: (1) Perception and Fusion, which uses pre-trained deep networks (e.g., CNNs or ViTs [6]) to process and integrate multi-modal sensor data; (2) Localization and Mapping, where BEV-based

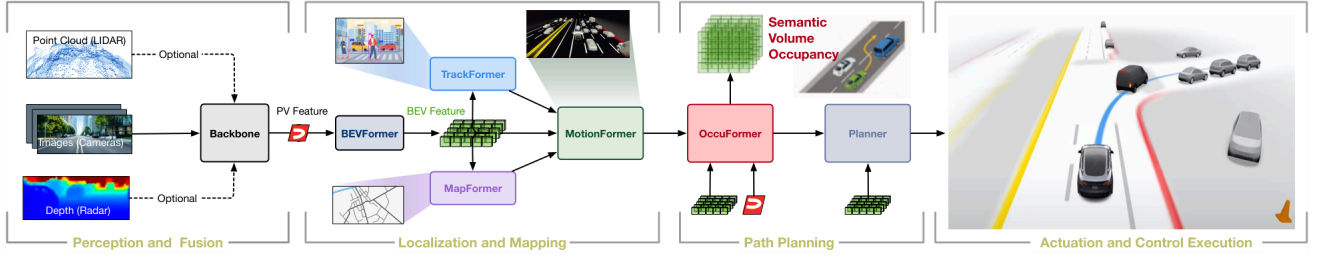


Fig. 4: Illustration of the Full Self-Driving (FSD) Model. A typical FSD system processes multi-source sensing data to construct a high-fidelity representation of the vehicle’s surroundings for navigation and decision-making.

modules (BEVFormer, TrackFormer, MapFormer) generate a centimeter-level 2D bird’s eye view (BEV) feature map centered around the vehicle, track other traffic participants, known as agents (e.g., vehicles, pedestrians), and identify static road elements; (3) Path Planning, where modules like OccuFormer and MotionFormer construct 3D occupancy maps and predict trajectories to plan safe driving routes; and (4) Actuation and Control, which translates planned maneuvers into vehicle control signals for vehicle’s actuator to execute.

Cameras are central to self-driving systems, with some companies (e.g., Tesla [7]) pursuing camera-only solutions. *Consequently, almost all FSD models process data frame by frame, synchronizing outputs from other sensors with the temporal unit of camera frames. In line with this practice, our FSC framework also operates on a per-frame basis.*

B. Hitchhiking I: Planned Trajectory

The first intermediate result utilized by our FSC framework is the vehicle’s *current position* at time t and its *planned trajectory* for the upcoming K frames. This planned trajectory is generated by the Planner component. Fig. 5 illustrates an example of such a trajectory, where the component predicts six positions over the next six frames. Formally, the planned trajectory is defined as:

$$\mathcal{P}_t = \{P_{t+1}^{\text{VEH}}, P_{t+2}^{\text{VEH}}, \dots, P_{t+K}^{\text{VEH}}\}, \quad (1)$$

where $P_t^{\text{VEH}}(x, y, z)$ represents the vehicle’s position at time t in geographical coordinates (north-south, east-west, altitude). The vehicle’s position is defined as its center. Assuming a constant relative displacement of the antenna array, denoted as ΔP^{ANT} , the antenna array positions are given by $P_t^{\text{VEH}} + \Delta P^{\text{ANT}}$. By leveraging the planned trajectory, our objective is to predict channel states at these future positions. This allows the communication system to proactively configure its beamforming coefficients in advance, ensuring optimal performance as the vehicle moves along its planned path. This approach eliminates the latency associated with measuring the channel through reference RF signals from satellites.

C. Hitchhiking II: Occupancy Map

The second intermediate result captured by our FSC framework is the *occupancy map* generated by the OccuFormer component of the FSD system. This occupancy map provides a semantic, volumetric representation of the space surrounding

the vehicle. The mapped area spans $51.2\text{ m} \times 51.2\text{ m} \times 8\text{ m}$, centered on the vehicle, and is divided into a grid of $256 \times 256 \times 40$ continuous, non-overlapping voxels. Each voxel is $20 \times 20 \times 20\text{ cm}^3$ in size and categorized into one of 17 common road-related classes, including ego vehicle, drivable surface, car, vegetation, barrier, and so on. Fig. 6 illustrates an example, where air voxels are rendered as transparent and other voxel types are color-coded.

Originally designed for collision avoidance and trajectory planning, we repurpose this map to model the vehicle’s radio frequency environment. By mapping voxel categories to their electromagnetic properties—conductivity σ and relative permittivity ϵ —using the ITU standard table [8], we compute the attenuation factor γ for each voxel in units of Np/m:

$$\gamma = \omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right)}, \quad (2)$$

where $\omega = 2\pi f$ is the angular frequency, μ is the permeability, and f is the signal frequency. Then, the attenuation coefficient $\alpha(V_t^i)$ caused by a voxel V_t^i is then given by:

$$\alpha(V_t^i) = e^{-\gamma(V_t^i)\Delta V} \cdot e^{\mathbf{J} \frac{2\pi\Delta V}{\lambda}}, \quad (3)$$

where λ is the wavelength, $\mathbf{J}$ denotes the imaginary unit, ΔV is the size of the voxel, and $\gamma(V_t^i)$ is the attenuation factor of the voxel. This formulation captures both the signal’s exponential decay and its phase shift due to propagation through the voxel.

Eventually, the occupancy map captured from the FSD model is updated as:

$$\mathcal{V}_t = \{V_t^1, V_t^2, \dots, V_t^N\} \text{ where } V_t^i = \langle P(V_t^i), \alpha(V_t^i) \rangle, \quad (4)$$

where $P(V_t^i)$ represents the voxel’s position in 3D space, and $\alpha(V_t^i)$ encapsulates its attenuation property. This representation enables accurate modeling of RF signal interactions with the environment.

D. Hitchhiking III: Envisioned Posture

The final intermediate result our FSC framework aims to capture is the predicted postures of the vehicle in the upcoming period. This information is critical for aligning voxels between consecutive occupancy maps (discussed later). Unfortunately, the current FSD model does not explicitly predict vehicle postures, as this is generally considered secondary for driving tasks. To overcome this limitation, we intercept the general

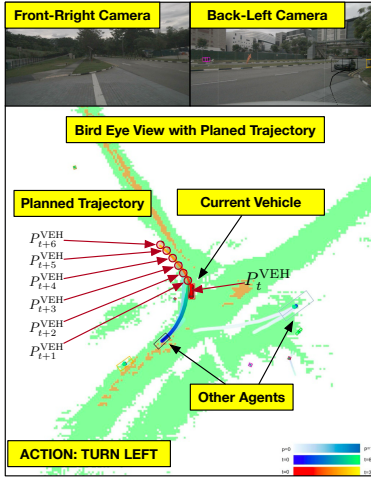


Fig. 5: Example of Planned Trajectory

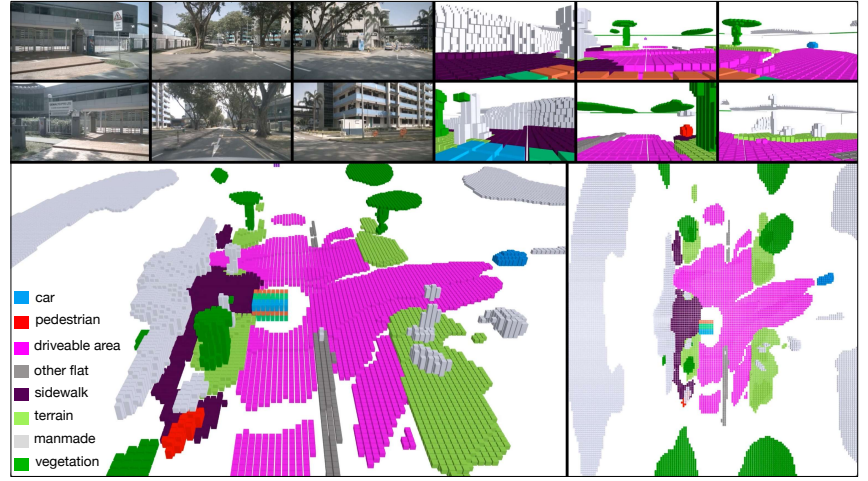


Fig. 6: Example of Semantic Occupancy Map

features outputted by the Backbone component of the FSD model and use them as keys and values in a cross-attention mechanism, with queries derived from the planned trajectory. This process is implemented using a Transformer decoder, termed PosFormer. Formally, the cross-attention is computed as:

$$\Psi_t \doteq \text{Softmax} \left(\frac{\mathcal{P}_t \mathcal{F}_t}{\sqrt{d}} \right) \mathcal{F}_t, \quad (5)$$

where $\mathcal{F}_t$ represents the captured features and d is a scaling factor. The output comprises K envisioned postures:

$$\Psi_t = \{\psi_{t+1}, \psi_{t+2}, \dots, \psi_{t+K}\}. \quad (6)$$

The design of PosFormer is motivated by the FSD system's ability to accurately estimate road smoothness and inclination by analyzing video streams from the vehicle's front-facing cameras. Specifically, it infers the vehicle's posture by leveraging the relative alignment between the road surface and surrounding structures.

V. LEARNING RADIO ENVIRONMENT

In this section, we detail how the intermediate results captured from the FSD are leveraged to model the radio environment using differentiable ray tracing.

A. Problem Definition

Our objective is to estimate the channel states, also known as channel impulse responses (CIRs), at the vehicle's planned trajectory over the upcoming K frames. Formally, these CIRs are represented as:

$$\mathcal{H}_t = \{h_{t+1}, h_{t+2}, \dots, h_{t+K}\}, \quad (7)$$

where h_{t+i} denotes the CIR at the predicted position P_{t+i}^{VEH} and time $t+i$, conditioned on the vehicle's current position P_t^{VEH} . At subcarrier frequency f , the CIR $h_t(f)$ is modeled as the superposition of the Line-of-Sight (LoS) propagation path and contributions from M multi-path components:

$$h_t(f) = h_t^0(f) + \sum_{i=1}^M h_t^i(f), \quad (8)$$

where $h_t^0(f)$ represents the LoS path and $h_t^i(f)$ corresponds to the i^{th} multi-path component. For simplicity, our analysis focuses on a single subcarrier, but the framework can be extended to all subcarriers by modeling them independently.

To compute the CIRs, we employ a differentiable ray tracing (DRT) algorithm to trace the RF signals from the satellite to the planned positions. This problem can be formulated as:

$$\text{DRT} \left(\underbrace{\widehat{\mathcal{P}}_t}_{\text{Trajectory}}, \underbrace{\Psi_t}_{\text{Posture}}, \underbrace{\widehat{\mathcal{V}}_t}_{\text{Map}}, \underbrace{\mathcal{O}_t}_{\text{Orbit}} \right) \rightarrow \underbrace{\widehat{\mathcal{H}}_t}_{\text{Channel}}, \quad (9)$$

where $\mathcal{O}_t = \{O_{t+1}, O_{t+2}, \dots, O_{t+K}\}$ denotes the satellite's orbit over the next K frames. Notably, the CIR at the vehicle's current position does not require prediction, as it can be directly estimated using the preambles of received packets or reference signals transmitted by the satellite.

B. Radiance Field

Before estimating the CIRs at the planned positions, we formally define the radiance field:

Definition 1 (Radiance Field) The radiance field $\mathcal{V}_t^+$ extends the occupancy map $\mathcal{V}_t$ by incorporating two additional radiance properties, $S(\mathcal{V}_t^i)$ and $R(\mathcal{V}_t^i)$, for each voxel $V_t^i \in \mathcal{V}_t^+$, beyond its position and attenuation factor. Here, $S(\mathcal{V}_t^i)$ represents the total RF radiance acquired through both direct and indirect propagation, while $R(\mathcal{V}_t^i)$ is a latent feature that captures the radiance directivity.

The radiance of a voxel is not unidirectional and varies with the receiving direction, resulting in an infinite number of potential radiating outcomes, which cannot be fully represented by a single variable $S(\mathcal{V}_t^i)$. To address this, we use a globally shared MLP denoted by $\Omega(\cdot)$ to learn the radiance patterns for each voxel based on a 32-dimensional learnable radiance feature $R(\mathcal{V}_t^i)$. The radiance from a voxel towards a specific direction ω is computed as follows:

$$S(\mathcal{V}_t^i, \omega) = \Omega(\omega, R(\mathcal{V}_t^i), S(\mathcal{V}_t^i)), \quad (10)$$

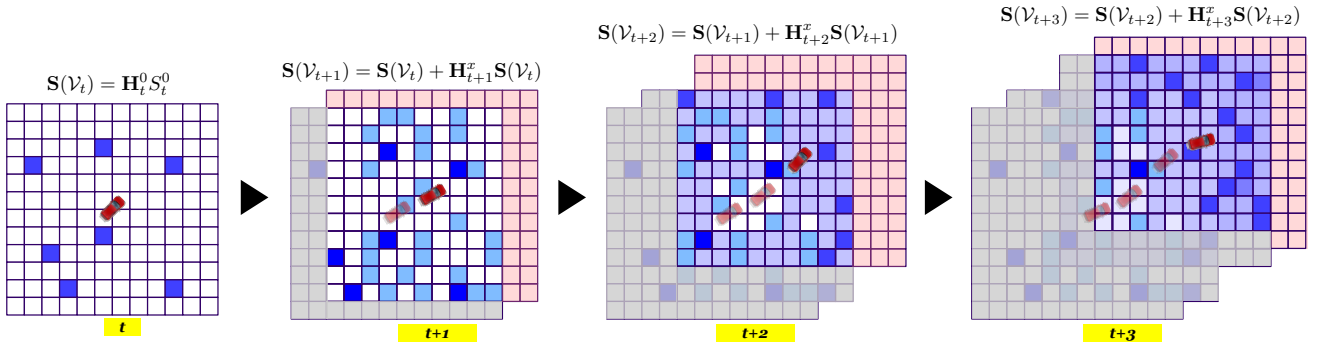


Fig. 7: Illustration of Progressive Ray Tracing. In each frame, FSC performs one tracing iteration based on the results from the previous frames. Initially, at frame t , only the LoS propagation is computed. As the vehicle moves forward, the first-order, second-order, and third-order reflections are incrementally incorporated at frames $t + 1$, $t + 2$, and $t + 3$, respectively, for the shared voxels. Outdated voxels in $(\mathcal{V}_{t+i}^+ - \mathcal{V}_{t+i+1}^+)$ (highlighted in gray) are discarded, while newly added voxels in $(\mathcal{V}_{t+i+1}^+ - \mathcal{V}_{t+i}^+)$ (highlighted in red) are initialized using LoS propagation.

A new radiance field $\mathcal{V}_t^+$ is constructed from the occupancy map $\mathcal{V}_t$ newly acquired in a frame. Each voxel $V_t^i \in \mathcal{V}_t^+$ is characterized by four key properties:

$$V_t^i \left(\underbrace{P(V_t^i), \alpha(V_t^i), R(V_t^i)}_{\text{Learnable}}, \underbrace{S(V_t^i)}_{\text{Traceable}} \right) \quad (11)$$

where $P(V_t^i)$ denotes the voxel's position, $\alpha(V_t^i)$ represents its attenuation factor, $R(V_t^i)$ is the radiance feature, and $S(V_t^i)$ is the radiance property. Notably, both $\alpha(V_t^i)$ and $R(V_t^i)$ are defined as learnable parameters, while $S(V_t^i)$ is traceable.

C. Progressive Ray Tracing

The RF signal $S(V_t^i)$ is iteratively updated through the following matrix equation, which models the complex interplay of direct and multi-path propagation:

$$\begin{bmatrix} S(V_t^1) \\ S(V_t^2) \\ \vdots \\ S(V_t^N) \end{bmatrix} = \begin{bmatrix} h_t^1 \\ h_t^2 \\ \vdots \\ h_t^N \end{bmatrix} S_t^0 + \begin{bmatrix} 0 & h_t^{1,2} & \dots & h_t^{1,N} \\ h_t^{2,1} & 0 & \dots & h_t^{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ h_t^{N,1} & h_t^{N,2} & \dots & 0 \end{bmatrix} \begin{bmatrix} S(V_t^1) \\ S(V_t^2) \\ \vdots \\ S(V_t^N) \end{bmatrix}, \quad (12)$$

or more concisely,

$$\mathbf{S}(\mathcal{V}_t) = \underbrace{\mathbf{H}_t^0 S_t^0}_{\text{LoS}} + \underbrace{\mathbf{H}_t^x \mathbf{S}(\mathcal{V}_t)}_{\text{NLoS}}, \quad (13)$$

where $\mathbf{S}(\mathcal{V}_t)$ represents the radiance distribution in the field. The S^0 represents the transmitted RF signal via the LoS propagation from the satellite. The CIRs $\{h_t^1, h_t^2, \dots, h_t^N\}$ model the LoS paths between the satellite and the i^{th} voxel. The h_t^i is estimated using the Friis formula if the path is unobstructed:

$$h_t^i = \sqrt{G_t G_r} \cdot \frac{\lambda}{4\pi d} \cdot e^{\mathbf{J} \frac{2\pi d}{\lambda}}, \quad (14)$$

where $d = \|O_t - P(V_t^i)\|$ is the distance between the satellite O_t and the voxel V_t^i , λ is the wavelength of the signal, G_t and G_r are the transmitter and receiver antenna gains, respectively. If the path is obstructed (e.g., due to occlusions), h_t^i is set to zero. The off-diagonal terms $h_t^{i,j}$ encode multi-

path interactions between voxels, capturing signal attenuation from voxel V_t^j to voxel V_t^i as:

$$h_t^{i,j} = \alpha(V_t^j) \cdot \prod_{k=1}^{W_{i,j}} \alpha(V_t^k), \quad (15)$$

where $W_{i,j}$ represents the number of intervening voxels between V_t^i and V_t^j , which cumulatively attenuate the signal. Thus, the reflected and/or scattered RF signals from voxel V_t^j to voxel V_t^i are computed as follows:

$$h_t^{i,j} S(V_t^j, \omega^{i,j}) = \left(\alpha(V_t^j) \cdot \prod_{k=1}^{W_{i,j}} \alpha(V_t^k) \right) \Omega(\omega^{i,j}, R(V_t^j), S(V_t^j)), \quad (16)$$

where

$$\omega^{i,j} = \frac{P(V_t^i) - P(V_t^j)}{\|P(V_t^i) - P(V_t^j)\|} \quad (17)$$

This formulation aligns with prior work in the field [1], [9].

At first glance, Eqn. 12 appears circular: the calculation of $S(V_t^i)$ depends on $S(V_t^j)$, and vice versa. Actually, it is a recursive equation that is progressively computed as follows:

$$\begin{aligned} \mathbf{S}(\mathcal{V}_t) = & \underbrace{\mathbf{H}_t^0 S_t^0}_{\text{LoS Prop.}} + \underbrace{\mathbf{H}_t^x (\mathbf{H}_t^0 S_t^0)}_{\text{1st-order Ref.}} \\ & + \underbrace{\mathbf{H}_t^x (\mathbf{H}_t^x (\mathbf{H}_t^0 S_t^0))}_{\text{2nd-order Ref.}} + \underbrace{\mathbf{H}_t^x (\mathbf{H}_t^x (\mathbf{H}_t^x (\mathbf{H}_t^0 S_t^0)))}_{\text{3rd-order Ref.}} + \dots \end{aligned} \quad (18)$$

Initially, the satellite illuminates voxels with unobstructed LoS paths. These illuminated voxels act as secondary radiance sources, propagating signals to other voxels (1st-order reflection). The process iterates indefinitely, inherently capturing arbitrary-order scattering effects. While more iterations improve accuracy, prior methods like NeRF² limit computations to two iterations akin to Eqn. 12 to reduce the prohibitive cost of handling millions of voxels.

We observe that adjacent occupancy maps (or radiance fields) between two frames are highly overlapping, enabling the reuse possibility of the previous frame's radiance field. Adopting a progressive approach, we compute a single iteration per frame, which is able to effectively accumulate infinite



Fig. 8: Data Collection Platform



Fig. 9: Example of Collected Data including Video and LiDAR Streams

iterations over time. Formally, Eqn. 18 is approximated as:

$$\begin{cases} \mathbf{S}(\mathcal{V}_t) \approx \mathbf{H}_t^0 \mathbf{S}_t^0 \text{ [LoS Prop.]}, \\ \mathbf{S}(\mathcal{V}_{t+1}) \approx \mathbf{S}(\mathcal{V}_t) + \mathbf{H}_t^x \mathbf{S}(\mathcal{V}_t) \text{ [1st-order Refl.]}, \\ \mathbf{S}(\mathcal{V}_{t+2}) \approx \mathbf{S}(\mathcal{V}_{t+1}) + \mathbf{H}_t^x \mathbf{S}(\mathcal{V}_{t+1}) \text{ [2nd-order Refl.]}, \\ \vdots \\ \mathbf{S}(\mathcal{V}_{t+n}) \approx \mathbf{S}(\mathcal{V}_{t+n-1}) + \mathbf{H}_t^x \mathbf{S}(\mathcal{V}_{t+n-1}) \text{ [nth-order Refl.]}. \end{cases} \quad (19)$$

When both the vehicle and the satellite are approximately stationary, $\mathbf{S}(\mathcal{V}_t) \approx \mathbf{S}(\mathcal{V}_{t+1}) \approx \dots \approx \mathbf{S}(\mathcal{V}_{t+n})$ and $\mathbf{S}_t^0 \approx \mathbf{S}_{t+i}^0$, the CIRs of $\mathbf{H}_t^0$ and $\mathbf{H}_t^x$ remain stable. As a result, $\mathbf{S}(\mathcal{V}_{t+n})$ aggregates n reflections, progressively approximating the ground truth over n iterations. Since only a single iteration is performed per frame, the computational load is significantly reduced. This approach enables the radiance field to iteratively converge toward the true propagation characteristics across successive frames. At 500 km/h and 2300 fps, the vehicle moves about 30.48 cm per frame, spanning at most two voxels. While satellite motion is rapid in orbit, it appears nearly static relative to the ground over short intervals, making the above approximation practical.

Fig. 7 illustrates this procedure. At frame t , we compute radiance for voxels with direct satellite LoS. Subsequent frames treat these voxels as sources to propagate signals to others. As the vehicle moves, shared voxels converge to their true radiance values. Voxels excluded from the new radiance field ($\mathcal{V}_{t+i}^+ - \mathcal{V}_{t+i+1}^+$) are discarded, while new voxels ($\mathcal{V}_{t+i+1}^+ - \mathcal{V}_{t+i}^+$) are initialized via LoS propagation. New voxels are initialized with updated satellite position O_{t+n} , ensuring long-term accommodation of satellite mobility.

After computing the radiance field, the CIRs $\mathcal{H}_t$ at the planned trajectory positions are queried and computed from the radiance field as follows:

$$\mathcal{H}_t = \left\{ \frac{S(V_t^1)}{S_t^0}, \frac{S(V_t^2)}{S_t^0}, \dots, \frac{S(V_t^K)}{S_t^0} \right\} \quad \text{s.t.} \quad P(V_t^k) = P_{t+k}^{\text{VEH}} + \Delta P^{\text{ANT}} \quad (20)$$

where V_t^k denotes the voxel where the antenna is located along the planned trajectory. P_{t+k}^{VEH} represents the planned position of the vehicle (i.e., the center of the vehicle) at the k^{th} frame, while ΔP^{ANT} is the constant displacement of the antenna relative to the vehicle's center.

D. Parameter Refining

The ray tracing algorithm relies heavily on several uncertain differentiable parameters (e.g., $\alpha(V_t^i)$, $R(V_t^i)$, $\Omega(\cdot)$). To address this issue, we employ the backward propagation to iteratively optimize these parameters whenever the commu-

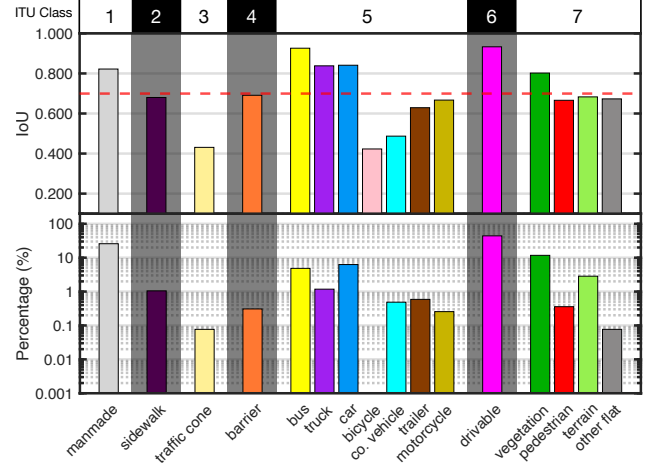


Fig. 10: Occupancy Map Prediction

nication system receives a reference signal from the satellite. The loss function for this optimization is defined as follows:

$$\mathcal{L} = \sum_{i=1}^{N_{\text{ANT}}} |S(V_t^i) - \hat{S}(V_t^i)|^2 \quad \text{s.t.} \quad P(V_t^i) = P_{t+i}^{\text{VEH}} + \Delta P^{\text{ANT}}, \quad (21)$$

where N_{ANT} is the number of the antenna elements equipped in the array, $S(V_t^i)$ represents the RF signal traced to the voxel accommodating the vehicle's antenna at time t , and $\hat{S}(V_t^i)$ denotes the actual received signal by the antenna. Theoretically, estimating $S(V_t^i)$ involves contributions from all voxels in the radiance field. This implies that the backpropagation process may update the parameters of all voxels, following the chain of dependencies below:

$$\mathcal{L} \rightarrow S(V_t^i) \rightarrow \begin{cases} \mathbf{H}_t^x \rightarrow \alpha(V_t^i); h_t^i \rightarrow (G_t, G_r) \\ S^0; S(V_{t-1}^i) \rightarrow \Omega(\cdot) \rightarrow R(V_t^i) \end{cases} \quad (22)$$

Particularly, as the vehicle moves to new positions, these parameters are gradually refined, leading to improved accuracy in the ray tracing model.

VI. IMPLEMENTATION

Existing self-driving datasets like nuScenes [10] lack communication-related data, prompting us to develop a custom platform. Fig. 8 shows our setup, which includes a vehicle equipped with six 10 fps cameras, a LiDAR, an IMU, a GNSS receiver, and a satellite antenna array for channel measurements. Data collection is managed by a high-performance laptop (Intel Core i9-13900HX, 16GB RAM, Ubuntu 20.04 LTS) using ROS bags. For RF signal acquisition, we used a 1 GHz SDR to capture data from GPS or Iridium satellites, as Starlink's communication protocols are proprietary and not

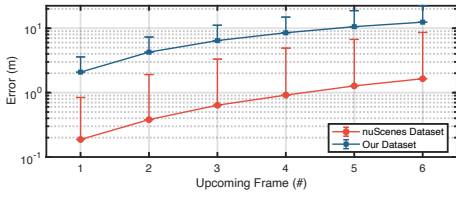


Fig. 11: Predicted Trajectory

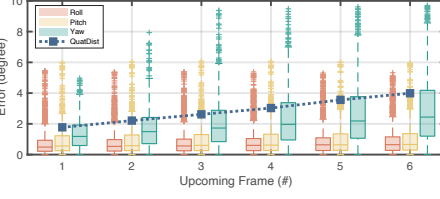


Fig. 12: Envisioned Posture

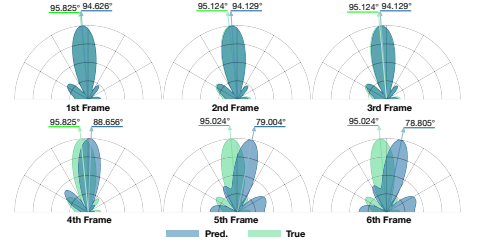


Fig. 13: Beamforming Effect

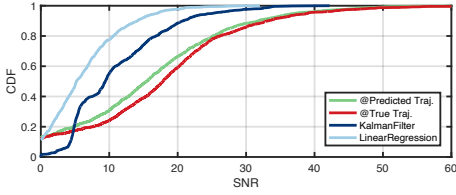


Fig. 14: Overall SNR

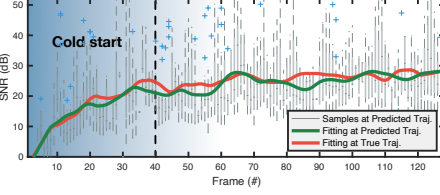


Fig. 15: SNR over Time

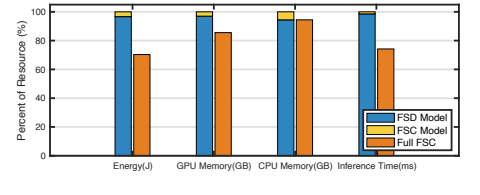


Fig. 16: Resource Consumption

publicly accessible. Fig. 9 illustrates an example of collection scene in an urban environment. Over a 20 km journey, we gathered 40,310 RF signals from 10 satellites across 202,600 GNSS-recorded locations, with vehicle speeds ranging from 0 km/h to 55 km/h, typical of urban driving.

We integrate UniAD [4] with OccFormer [5] to form the FSD model. UniAD provides trajectory planning and posture prediction, while OccFormer generates a semantic 3D occupancy map with $256 \times 256 \times 40$ spanning $51.2 \times 51.2 \times 8m^3$. Both models are pre-trained on the nuScenes dataset [10], which contains 1000 driving sequences annotated at 2 Hz. For the PosFormer, we adopt a lightweight Transformer Decoder that consists of a 3-layer decoder with four attention heads and 64-dimensional embeddings, striking a balance between model capacity and the risk of overfitting, given the relatively small input size. It is trained by using IMU-derived true postures from the dataset. For the MLP $\Omega(\cdot)$, we adopt a shallow MLP consisting of two hidden layers (64 neurons each, ReLU activation) followed by an output layer with two components: one neuron with a sigmoid activation to produce the amplitude (normalized to $[0, 1]$) and another neuron with a tanh activation to produce the phase (normalized to $[0, 2\pi]$). These outputs are combined to form the complex-valued RF radiance. For experiments, we apply the pre-trained FSD model to our new dataset without further training. The FSD's intermediate outputs are fed into our FSC model to predict channel states for the next six frames. Optimization is performed on an NVIDIA GeForce RTX 4090 GPU (24GB) hosted on an AMD Ryzen Threadripper PRO 16-Core workstation.

VII. EVALUATION

A. FSD Microbenchmark

First, we evaluate the performance of the FSD in our new scenario. Its performance is critical as it directly impacts the effectiveness of our FSC. We assess the FSD from the following three perspectives:

(1) Trajectory Prediction: Fig. 11 illustrates the accuracy of trajectory prediction, implemented using UniAD. The trajectory prediction was evaluated against GNSS ground truth over six upcoming frames in nuScenes and our dataset. Errors grow over time due to prediction accumulation, but the mean errors in the first two frames remain acceptable (nuScenes: 0.1872 m, 0.3807 m; ours: 2.0819 m, 4.2524 m), supporting reliable short-term channel prediction and adaptation.

(2) Occupancy Prediction: Fig. 10 (top) shows the performance of the 3D semantic occupancy network, implemented using OccFormer [5]. Evaluated on the nuScenes dataset (due to the lack of ground truth in our dataset), it achieves an overall mean accuracy of 69.95%, measured using mIoU across material classes. The mIoU for specific ITU classes are: 82.20% (Concrete), 68.00% (Brick), 43.10% (Plasterboard), 69.10% (Wood), 68.73% (Metal), 93.30% (Very Dry Ground), and 70.60% (Medium Dry Ground). The low accuracy for plasterboard, primarily due to traffic cones, has minimal impact on RF propagation.

Fig. 10 (bottom) also summarizes the percentage of non-air voxels in our scene, which includes 16 non-air classes (excluding motorcycles). Air voxels account for 99.61% of the total, confirming the sparsity of the maps.

(3) Posture Prediction: We finally evaluate the accuracy of the proposed PosFormer. The posture is represented in Euler angles. The mean angle errors for the first three frames are $(0.696^\circ, 0.798^\circ, 1.422^\circ)$, $(0.725^\circ, 0.825^\circ, 1.919^\circ)$, and $(0.761^\circ, 0.852^\circ, 2.356^\circ)$, as shown in Fig. 12. Notably, the error increases over time because the queries of PosFormer depend on the predicted positions, whose accuracy degrades with time. While posture errors may introduce inaccuracies when aligning voxels in adjacent occupancy maps.

In short, the existing FSD model still suffers from inaccuracies, particularly in the planned results over time. This limitation may impact the overall performance of FSD. To address this, we strive to leverage the most recent prediction results for the FSD system.

B. End-to-End Performance of FSC

Next, we evaluate the end-to-end performance of FSC in terms of its channel prediction capability. To quantify the accuracy, we adopt the SNR metric (as used in [11] and [1]) as follows:

$$\text{SNR} = \log_{10} \frac{\|\mathbf{H}_{\text{gt}}\|^2}{\|\mathbf{H} - \mathbf{H}_{\text{gt}}\|^2} \quad (23)$$

where the difference between the predicted and ground truth CIRs is treated as noise. A higher positive SNR indicates better alignment with the ground truth, while a negative SNR suggests that the noise dominates the signal.

(1) Overall SNR. We evaluate the predicted SNR at the next position after each move and compare FSC with two traditional methods: linear regression and Kalman Filter, both of which rely solely on historical observations. As shown in Fig. 14, the mean SNRs of linear regression and Kalman Filter at the true next positions are 6.218 dB and 11.041 dB, respectively. In contrast, FSC achieves mean SNRs of 17.264 dB at the true next position and 15.453 dB at the predicted position, outperforming the baselines by 11.046 dB and 6.223 dB. These results demonstrate that FSC benefits from leveraging spatial information via the occupancy map and jointly modeling trajectory prediction and channel estimation, enabling robust performance under dynamic environmental changes where traditional methods fail.

(2) Beamforming Effect. We evaluate the use of predicted CIRs for beamforming by comparing results obtained from predicted and measured CIRs at true future positions. Fig. 13 shows beamforming results for the next six frames using the classical Bartlett algorithm [12]. The blue and green lobes denote the power distributions of predicted and measured CIRs, respectively. The angular errors between their main lobes are 0.995° , 0.995° , 1.199° , 7.169° , 16.02° , and 16.219° from the first to sixth frames, consistent with the SNR analysis and indicating that the first two or three predicted positions are most reliable for communication planning.

(3) Cold Start. FSC employs a progressive ray tracing algorithm to refine the radiance field estimate as the vehicle moves. To observe the changes, we plot the predicted SNR results as a function of the frame in Fig. 15. The grey boxes represent predicted SNR samples at the predicted position (fitted by the green line) and the true position (fitted by the red line). The results indicate a short cold start phenomenon, where our FSC requires about 40 frames to stabilize the SNR, regardless of using true or predicted positions. This occurs as the model depends on more RF measurements collected at different positions to achieve convergence to the optimal parameters.

(4) Resource Consumption. Finally, we measure the resource consumption of the framework using the NVIDIA System Management Interface (nvidia-smi) tool. The results are shown in Fig. 16. FSC consumes only 3.35%, 3.01%, 5.58%, and 1.45% of the total energy (J), GPU memory (GB), CPU memory (GB), and inference time (ms), respectively, due to the hitchhiking approach. In contrast, the majority of

resources are consumed by running the FSD model. We also provide a rough estimation of the resource consumption for a full FSC, which operates from scratch without reusing the intermediate results of the FSD model. Approximately 70.32% energy, 85.56% GPU memory, 94.42% CPU memory, and 74.21% time are required compared to the hitchhiking method. This demonstrates the efficiency of the hitchhiking approach.

VIII. RELATED WORK

Our work is related to the following two primary fields;

- **Self-Driving Intelligence:** Autonomous vehicles integrate sensing, planning, and control to operate safely and efficiently with minimal human intervention. The Society of Automotive Engineers (SAE) defines six automation levels from Level 0 to Level 5 [13]–[16], and industry leaders such as Waymo, Cruise, Baidu, and Tesla are actively advancing toward full self-driving (FSD). Recent research has driven substantial progress across perception, prediction, and planning, including self-supervised LiDAR motion learning [17], planning-centered frameworks [4], robust representation learning [18], collaborative perception [19], vectorized scene modeling [20], multi-agent multimodal datasets [21], cross-dataset semantic learning [22], multi-future trajectory prediction [23], realistic video simulation [24], and full-scale monocular depth estimation [25], collectively accelerating the development of reliable autonomous driving systems.

- **Channel Prediction:** Traditional methods rely on CSI estimation and feedback [26], [27], often hindered by delays and errors. Solutions include robust power allocation [28], [29], NOMA [30], [31], and spectrum-sharing techniques [32], [33]. Training sequence-based methods [32]–[34] and advanced approaches like maximum likelihood [35] and MAP [36], [37] have enhanced robustness, while dynamic tracking uses Kalman filtering and Markov processes [38]–[41]. Deep learning has emerged as a powerful alternative, applied to cross-band prediction [42], UL/DL calibration in massive MIMO [43], and intelligent surface-based path deployment [44]. Models like NeRF2 [1] and NeWRF [45] leverage WRF reconstruction for radio wave modeling.

IX. CONCLUSION

This work introduces an efficient and safe framework for radio radiance field modeling in self-driving systems, combining hitchhiking execution and progressive ray tracing. This work advances robust vehicular communication systems for autonomous driving, with future research focusing on real-world optimization and broader applications.

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